ACSM Submission Tool v.1.0

Installation

To load the ipf procedure within Igor, the user can:

- Place the file in the Igor Procedure folder, to automatically loaded at the startup of Igor
- Double click on the file

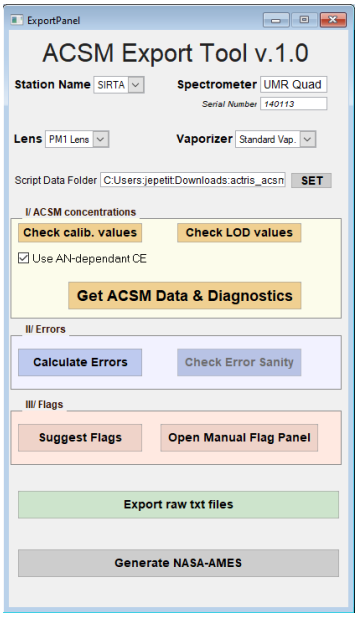
This tool is fully compatible with Igor 6.37.

Setup

Requirements

- ACSM data need to be loaded within Igor. It is highly preferable to select periods with constant IE/RIE values.
- Data need to be processed using all acsm_local tools
- User should avoid using the Blacklist tool. Flags are prefered instead.

The panel can be loaded using PopUp Menu “ACMCC Annual Submission”.



- Set station name : The drop down menu contains a list of different stations. Station name is only indicative, it is not used for metadata, only for filenaming.
- Procedure automatically detects type of ACSM, based on folders present in the data browser. In the case of Q-ACSM, serial number is automatically detected from DAQ_Matrix. In the case of ToF-ACSM, serial number shall be entered manually.
- Lens & vaporizer type can be set using corresponding dropdown menus.
- Script Data folder is the folder containing the Python scripts for NASA-AMES formatting.

ACSM Concentrations

The user needs to check first the calibration & LOD values. IE & RIE values in Review batch needs to be explicitely checked.

In the case of standard vaporizer, an ammonium nitrate dependent CE correction can be applied, following Middlebrook et al. (2012). Note that in the case of capture vaporizer, this correction is disabled, assuming a CE of 1.

Pressing "Get ACSM data & diagnostics" automatically gathers variables in a specific data folder. At this stage of the development of the tool, the import of pump and dryer data is disabled.

Errors

Error matrices for each specie are calculated following Allan et al. (2003). Integrated error is the quadratic sum of mz's for each specie.

Flags

Pressing suggest flags triggers an automatic procedure to go through the data, and validate/invalidate rows following technical criteria (InletP variation, Airbeam, Vaporizer, LOD, concentration peaks). Corresponding flags are applied automatically. Note that, at this stage, these flags are only suggestions. The user has the possibility to overwrite all flags on the manual flag panel.

Clicking "Open Manual Flag" creates a panel with the timeserie of concentrations.



Automatic flags are added in the background.

Invalid flags are RED; warning (but valid) flags are grey. For clarity, and graphical purposes only, the user can hide invalid data (they are not removed from the dataset).

To apply a flag, the user needs to draw a rectangle (left click on the graph). The rectangle dimensions that matter are over the x-axis (not the y-axis). Once drawn, select the flag from the list, and on which specie to apply the flag. Then click “apply & update”.

Export raw text files

Once flagging has been properly completed, the user can click on “Export rax txt files”. This will create two files (concentration & diagnostics, and flag) in the script folder (under /data/StationName/in/).

Generate NASA-AMES

Requirements

Before clicking the button, make sure metadata & code have been correctly setup (see dedicated manual).